

DRY-RUN GUIDE | palindrome2.py
Source: Python\2nd_Belt\palindrome2.py

CORE BEHAVIOR

Compares a value/string with its reverse to check palindromicity.

REPRESENTATIVE INPUT

'level'.

STEP TRACE

Reverse is 'level'; equality succeeds.

EXPECTED OUTPUT / RETURN

True/palindrome message as specified by source.

ACTUAL SOURCE EXCERPT

Verbatim excerpt from Python\2nd_Belt\palindrome2.py; line numbers refer to the original file. No source was modified.

001 | # Question: Check whether a string or number is a palindrome.

002 | # Add your solution here.

003 |

004 | num = int(input("Enter the number:"))

005 |

006 | def is_palindrome_math(num: int) -> bool:

007 |

008 | if num < 0 or (num % 10 == 0 and num != 0):

009 | reversed_num=0

010 | m=num

011 | n=reversed_num

012 |

013 | while num > 0:

014 | digit = num % 10

015 | reversed_num = (reversed_num * 10) + digit

016 | num //= 10

017 |

018 | if n == m:

019 | print(f"The number {num} is a palindrome.")

020 | else:

021 | print(f"The number {num} is not a palindrome.")

022 |

023 |

DRY-RUN NOTE

Values are representative. For function-only modules the guide uses an illustrative direct call and explains its return; quirks/errors describe actual behavior.